

There's no place like home

(residence of electronics enthusiasts)

Mission

About

3.1.3. Full adder experiment

Components:

- breadboard
- 5V power supply
- 5mm red and green LED
- 3 x SPST pushbutton switches
- 2 x 220Ω , 3 x $2.2K\Omega$ resistors
- 7486 XOR, 7408 AND, 7432 OR chip with configuration:
 - Pin 14 as VCC, Pin 7 as ground
 - the rest is self-explanatory from the schematic

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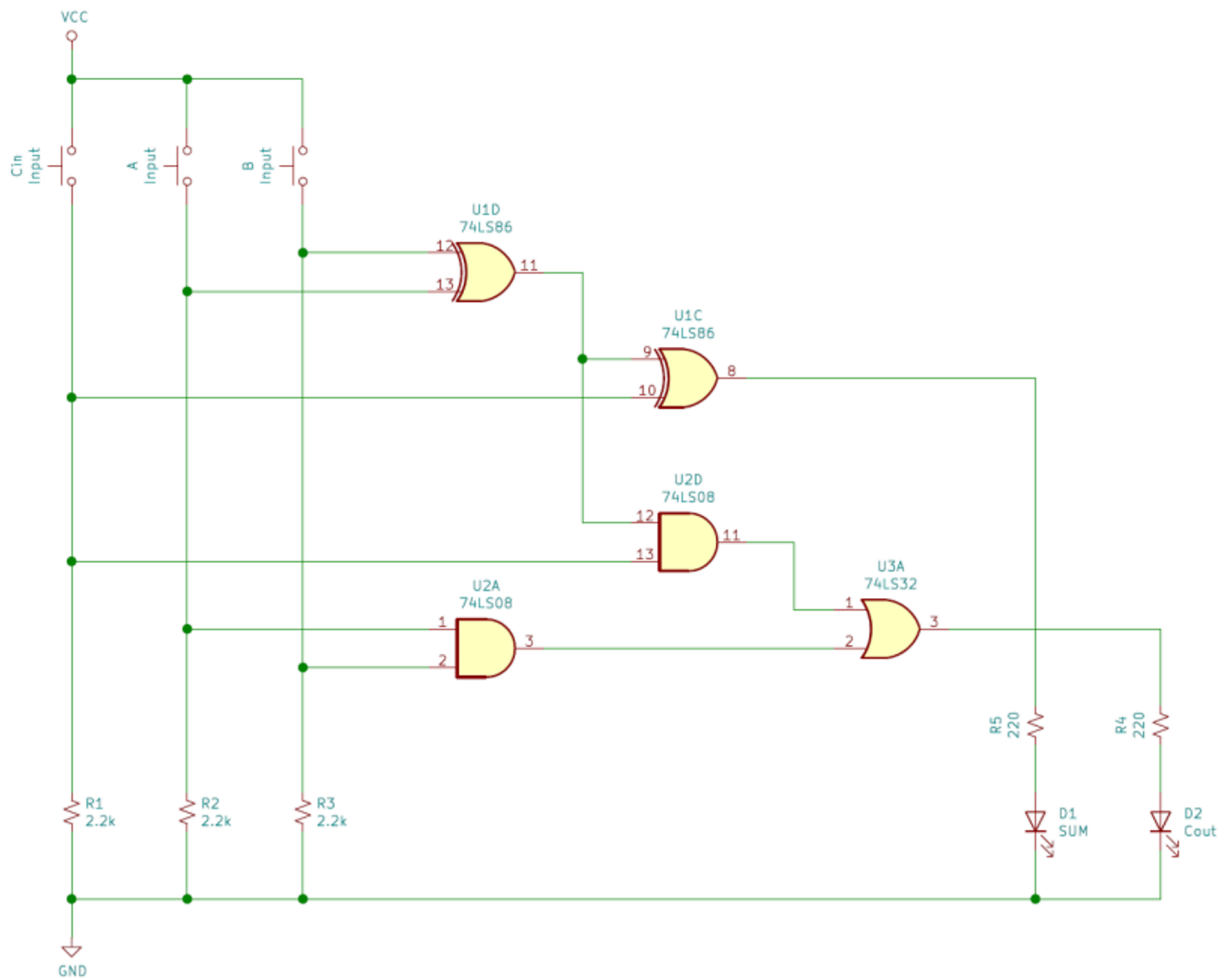


Figure 3-3 Full adder schematic

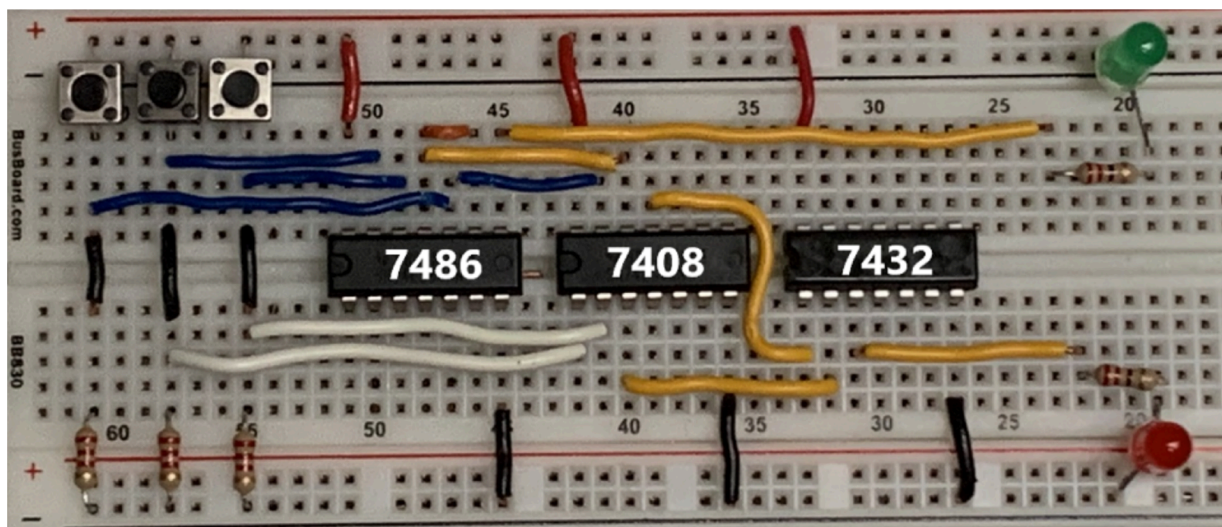


Figure 3-4 Full adder circuit

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scenarios:

- if inputs A and B are different, the output of U1D is 1, so Cin must be 0
- if inputs A and B are the same, the output of U1D is 0, so Cin must be 1

Further, the green D2 diode represents carry, and it is lit only if any of the inputs of U3A is 1, in the following scenarios:

- if inputs A and B are both 1, the output of U2A is 1
- if inputs A and B are different, the output of U1D is 1, which makes one of the inputs of U2D value 1. Cin then also needs to be 1, because it makes the second input of U2D

Once we have realized the sum of the discrete components, we can use the 7483 chip which performs binary addition of two 4-bit words. This rounds up our paradigm which should be clear.

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